

Chen Yi-Heng

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EDUCATION

Nanyang Technological University, Singapore

Aug 2024 - Jun 2028

Bachelor of Engineering in Mechanical Engineering with a Second Major in Business, cGPA: 4.71/5.00

- Dean's List AY24/25, School of Mechanical and Aerospace Engineering
- Dean's List AY25/26, School of Mechanical and Aerospace Engineering

EXPERIENCE

Archimedes Autonomous Vehicles, RobotX 2026 Team

May 2026 – Present

Member, Mechanical

- Designed adjustable mounts for sensors and electronics utilizing SolidWorks
- Replicated waterproof electronics box with waterproof penetrators on SolidWorks, creating a layout and related fixture points in preparation of assembly and simulation
- Executed active prototyping and fabrication of mounting components and structural supports

Undergraduate Research Experience on CAMPUS (URECA)

Aug 2025 – June 2026

Student Researcher

- Designed a fixture using SolidWorks to secure test subjects to the Shimadzu AG-X Plus 10kN test machine
- Conducted literature review for testing prosthetic sockets under ISO 10328 standards
- Devised and used experiment procedures for ultimate load of test subjects under axial compression
- Prepared strain gauges to quantify deformation of test subjects for future experiments
- Developed a research poster and wrote a research report highlighting findings and suggestions for future research

Engineering Innovation Design Project, GrubGrab (2nd Place, Safety & Health)

Jan 2026 - May 2026

Team Lead

- Led a team of 9 engineering students in conceptualisation and development of a mud-clearing robot for disaster relief
- Executed electronics design using Arduino, motors, drivers, and sensors
- Oversaw two departments in structural design of vehicle body and belt-conveyor system utilising SolidWorks
- Provided FDM 3D Printing support to fabricate customised beams, sprockets, and mounts for rapid prototyping
- Delegated manufacturing workload with milling machines and lathes and led active troubleshooting of printed parts
- Devised an intuitive 5-channel controlled user interface with Arduino C

Machine Element Design

Jan 2026 - May 2026

Team Lead

- Executed calculations of power, forces, and moments for element choices of a sugar cane juicing machine; proposed solutions for a 3-stage belt-gear-gear speed reduction transmission
- Devised radii for 2 shafts for the power transmission system based on calculated force elements, referring to industry catalogues to select bearings and retaining rings
- Wrote and refined a 40-page report discussing detailed decisions made in designing the transmission.

CO-CURRICULAR ACTIVITIES

Socials Sub-committee, NTU Crescent Hall

Oct 2025 - Present

Member

- Designed and planned an event of 12 games and crafts with a team of 8 peers
- Regulated games and implemented activities for 4 events, hosting up to a total of 300+ participants

Orientation Committee, MAE Club

Oct 2024 - Aug 2025

Logistics Main-committee Member

- Managed and implemented end-to-end logistics and on-site programme flow in a 6-member team for freshmen orientation, an event with 500 participants
- Optimised the organisation and assembly of 500+ welfare packs of 20+ items

SKILLS

- Languages: English (fluent), Mandarin (native)
- Technical Skills: Engineering drawing, Solidworks, Ansys Workbench, Ansys Fluent, Python, Tableau, Arduino, 3D printing, Strain gauging, Basic machining (turning, milling, etc.)